

Geonyeong Son

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RESEARCH INTERESTS

Structuring knowledge and experience curation, developing human-like language understanding, providing interactive learning and teaching, advancing maturity-inspired AI models and techniques.

EDUCATION

- **Hanyang University** Mar 2024 – Feb 2026
 - M.S. in Data Science Seoul, South Korea
- **Sejong University** Mar 2023 – Feb 2024
 - M.S. in Data Science Seoul, South Korea
- **Sejong University** Mar 2017 – Feb 2023
 - B.S. in Data Science Seoul, South Korea
 - Leave of absence for military service: Dec 2017 – Aug 2019

RESEARCH EXPERIENCE

- **Hanyang University** Mar 2024 – Feb 2026
Graduate Research Assistant (Advisor: Prof. Misuk Kim) Seoul, South Korea
 - Designed a distillation framework that updates the teacher using feedback from high- and low-capacity students before final distillation, improving accuracy by approximately 6.86% on average over other KD methods on five commonsense reasoning tasks including QASC and ARC-Easy.
 - Integrated query-specific granular and cluster-derived holistic instructions, improving accuracy by 29.1% on average over baselines on five commonsense reasoning tasks including CODAH and PIQA.
 - Built WHY-Chain from consecutive why-questions and refined three language models with span- and sentence-level objectives, improving accuracy by approximately 11.22% on average over baselines on four commonsense reasoning tasks including COPA and CSQA.
- **Sejong University** Mar 2023 – Feb 2024
Graduate Research Assistant (Advisor: Prof. Misuk Kim) Seoul, South Korea
 - Built a two-stage dialogue model using BERT-series encoders for commonsense retrieval and dual BART encoders for embedding fusion, improving performance by 20.85% on average across METEOR, ROUGE-L, and BERTScore metrics on PERSONA-CHAT.
 - Manually labeled over 100,000 sentences from Korean sustainability reports for the ESG-Kor dataset.
 - Authored the table-to-text and applications sections of a survey on data-transformation strategies for removing format heterogeneity.
- **Sejong University** Jul 2021 – Feb 2023
Undergraduate Research Assistant (Advisor: Prof. Misuk Kim) Seoul, South Korea
 - Built a dual-encoder KoBART dialogue generator with FAISS-based commonsense retrieval and embedding averaging, achieving a 15.6% BLEU improvement on Korean Chatbot data.

PROFESSIONAL EXPERIENCE

- **Gravylab** Apr 2026 – Present
Associate Researcher Seoul, South Korea
 - Architecting a novel data-driven recipe generation and menu planning framework by integrating diverse ingredient substitutions, and nutrition calculation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT

International Publications

- [J.3] [G.Son](#) & [M.Kim](#). (2026). **RTKD: Responsive Teacher Knowledge Distillation with Heterogeneous Students**. *Knowledge-Based Systems*, Vol. 345, 116101. DOI: 10.1016/j.knosys.2026.116101. (Q1, Impact Factor: 7.2).
- [C.2] [G.Son](#), [J.Lee](#), [M.Kim](#). (2025). **From Curiosity to Clarity: Exploring the Impact of Consecutive Why-Questions**. In Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, NM.
- [J.2] [G.Son](#) & [M.Kim](#). (2025). **Multi-Level Prompting: Enhancing Model Performance through Hierarchical Instruction Integration**. *Knowledge-Based Systems*, Vol. 320, 113545. DOI: 10.1016/j.knosys.2025.113545. (Q1, Impact Factor: 7.2).

[C.1] J.Lee, G.Son, M.Kim. (2024). **ESG-Kor: A Korean Dataset for ESG-Related Information Extraction and Practical Use Cases**. In Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, FL.

[J.1] G.Son & M.Kim. (2024). **A Simple and Efficient Dialogue Generation Model Incorporating Commonsense Knowledge**. *Expert Systems with Applications*, Vol. 249, Part B, 123584. DOI: 10.1016/j.eswa.2024.123584. (Q1, Impact Factor: 8.5).

Preprint

[P.1] S.Yoo, J.Lee, C.Yoon, G.Son, H.Hong, S.Seo, S.Yim, C.Jung, J.Park, M.Kim, and Y.Jang. (2025). **Data Transformation Strategies to Remove Heterogeneity**.

PROJECTS

- **Smart Manufacturing Technology for Customized Foods** Apr 2026 – Present
Gravylab MAFRA | Government Project
 - Architecting a novel data-driven recipe generation and menu planning framework by integrating diverse ingredient substitutions, and nutrition calculation.
- **Implementation of cloud-based no-code environment front-end tool for SW global market entry in Manufacturing/Finance/Service sectors** Apr 2024 – Feb 2026
Hanyang University IITP | Public-Sector Project
 - Developed a query-intent classifier that categorized user requests into graph, table, or combined outputs.
- **Development of Questions and Answers based on Insurance Product Documents** Apr 2024 – Oct 2024
Hanyang University Hanwha General Insurance Co., Ltd. | Corporate Project
 - Developed an adaptive RAG framework that selected chunking, embedding, and retrieval strategies for each QA.
- **K-ESG-based performance management and production process optimization service technology development for small and medium-sized manufacturing industries** Mar 2024 – Dec 2024
Hanyang University KEIT | Public-Sector Project
 - Constructed a Korean ESG dataset via human labeling in accordance with global and domestic ESG standards.
- **Development of a commonsense-aware and emotion-sensitive human-like dialogue system** Jun 2023 – Jun 2025
Sejong University / Hanyang University NRF | Government Project
 - Incorporated commonsense sentence embeddings into dialogue generation to improve response quality.
- **Development of heterogeneous fusion data detection and tracking technologies** Nov 2021 – Dec 2023
Sejong University IITP | Public-Sector Project
 - Developed a multi-modal fake-news detection framework that fused news text and images.

AWARDS AND HONORS

- **Excellence Award, Undergraduate Student Paper Competition, KCC 2022** Jun 2022
Korean Institute of Information Scientists and Engineers (KIISE)
 - Paper: [Dialogue Generator with Commonsense based on KoBART](#)

TEACHING EXPERIENCE

- **Teaching Assistant** Mar 2024 – Jun 2024
Hanyang University (Dept. of Data Science)
 - Machine Learning [Course Code: 12096, 13190] (Instructor: [Prof. Misuk Kim](#))
 - Led full-semester practical labs.

SKILLS

- **Programming Languages:** Python, Pytorch, Tensorflow, C, R, SQL, C++, Java

REFERENCES

Misuk Kim Associate Professor Data Science Hanyang University Email: misukim@hanyang.ac.kr Relationship: [B.S. & M.S. Advisor]	Pilsung Kang Professor Industrial Engineering Seoul National University Email: pilsung_kang@snu.ac.kr Relationship: [Reference]	Jaewook Song Associate Professor Industrial Engineering Hanyang University Email: jwsong@hanyang.ac.kr Relationship: [Reference]
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